

Advanced Self-Improving Ramp Metering Algorithm based on Multi-Agent Deep Reinforcement Learning

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Abstract—We proposed a novel ramp metering algorithm embedding multi-agent deep reinforcement learning (DRL) techniques, based on the data of loop detectors. A multi-agent DRL framework is adopted to generate proper ramp metering scheme for each ramp meter in real time to improve the operation efficiency of urban freeway with less investment. A simulation platform is developed to simplify the implementation and training of the algorithm. A set of simulation experiments – encompassing both single and multi-ramp scenarios with various traffic demand profiles – are conducted. Comparing with the state-of-the-practice ramp metering methods, the simulation results demonstrate that the proposed DRL-based algorithm outperforms in a comprehensive evaluation index considering mainstream speed at bottleneck and queue size on ramp. The method presents robustness, scalability, and the capability of further improvement by online learning during implementation.

Index Terms—ramp metering, deep reinforcement learning, adaptive control.

I. INTRODUCTION

Urban freeway takes a significant role in providing efficient personal mobility and facilitating freight movement in metropolitan districts. Effective traffic management on urban freeway would provide tremendous economic, security, and environmental benefits. The available strategies for traffic management on urban freeway include ramp metering and variable speed limits. In this work, we study the ramp metering approach that control the amount of flow entering the freeway through on-ramps, thus ameliorate traffic conditions in the mainline of freeway.

A. Review on Ramp Metering Methods

Over the last decades, efforts have been done to develop ramp metering algorithms which are capable of alleviating congestion and resultant delays more effectively. The existing ramp metering strategies can be categorized into the following three types: fixed-time strategies, adaptive ramp-metering strategies and model predictive control strategies.

Fixed-time strategies refer to the off-line metering measures for particular times-of-day. Given the constant historical demands, the ramp metering problem can be formulated as a linear programming or quadratic programming problem. The objective criterion is to maximize the number of served vehicles, or to maximize the total

traveled distance. Constraints may incorporate mainstream flow and ramp queue length restrictions. Most early-stage ramp metering strategies fall into the category.

Adaptive or traffic responsive ramp metering strategies use real-time measurements to capture the dynamic traffic demand, then variable metering rates are allocated to ramps in response to actual traffic conditions. The objective of the strategies is to keep the freeway traffic conditions close to prespecified values. The most widely deployed adaptive ramp metering strategies are ALINEA [1] and its variants. METALINE [2] is an extension of ALINEA for coordinated control of ramp meters. Other examples include Zone [3], Bottleneck [4], and fuzzy logic algorithm [5].

Model predictive control (MPC) strategies [6] adopt a freeway traffic model to predict the future traffic behavior in a rolling horizon framework, then optimize the ramp metering rates over the prediction horizon. The upside of MPC is the proactive feature, i.e. the pre-consideration of traffic condition in the near future. Several recent studies, e.g. [7], evaluated the controllers in simulation frameworks.

Despite many successful applications, the existing ramp metering options are not without drawbacks. Most existing strategies predefine traffic models to predict traffic patterns and calculate control actions, while the models probably fail in unusual conditions, e.g. extreme weather and traffic incidents. Moreover, administrators have to take effort to fine-tune the parameters of control algorithms. In addition, the existing strategies have no capability to improve the control efficiency in long-term implementation.

B. Reinforcement Learning and Its Application

Recent years have witnessed the rapid development of reinforcement learning (RL). RL refers to an artificial intelligence technique in which agents interact with the environment, learn an optimal policy by trial-and-error, then make sequential decision to solve a specific problem. It is a powerful tool for control and has demonstrated success in many complex problem settings such as video games [8], robot control [9], self-driving [10], chess and go [11]. In most cases, RL requires no explicit model of the environment beforehand and learns only from the past experiences. That is, RL works in a model-free fashion. (There are actually model-based RL algorithms, but they will not be discussed in this paper.) Also, RL has the ability to adapt different environments without change of algorithm and improve performance by continuous self-learning. For these reasons, RL is promising to overcome the downsides of the traditional ramp metering strategies. Application of RL to adaptive traffic signal control of intersections has been proven to be efficient

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in literatures [12]–[14]. It should be convincing that the paradigm can produce effect in ramp metering problem.

C. The Contribution of the Paper

In this paper, a novel ramp metering algorithm based on multi-agent deep reinforcement learning (DRL) framework is proposed. Multi-Agent Proximal Policy Optimization (MAPPO) architecture is introduced to solve ramp metering problem. A ramp metering testbed is developed based on SUMO microscopic traffic simulation software, providing a simulation environment encompassing several ramp metering scenarios, which is similar to OpenAI Gym [15], to process the interaction between agent and freeway environment, then evaluate the performance of RL-based ramp metering algorithms. Simulation experiments are undertaken to illustrate the power of the proposed approach. Superior performance compared to the baseline is presented.

II. PROPOSED ALGORITHM

An advanced self-improving ramp metering algorithm is developed for agents to learn how to execute on-ramp metering actions in a freeway environment. The following section gives a description of the RL-based ramp metering algorithm.

A. State, Action and Reward Representation

1) State

According to the classical traffic flow theory, characteristics of traffic state on the freeway can be depicted by three fundamental quantities, i.e., volume, density and speed. In the practice of traffic engineering, occupancy which is proportional to density is directly collected by the inductive loop detectors. Hence, the occupancy should be a natural and interpretable state representation of freeway traffic environment. Compared to other traffic state detectors such as millimeter wave radar and video camera, loop detector is more adaptable (e.g., less sensitive to weather conditions) and low-budget (pre-facilitated in most freeways).

To configure a ramp metering system, the freeway can be divided into several segments. Each segment S_i encompasses an entrance ramp R_i facilitated with a ramp meter (Fig. 1). The instantaneous traffic state of a segment will be detected by a set of mainstream loop detectors. Define the occupancy of detection profile k in segment i as $\bar{o}_{l,m,k,t}^{(i)}$, then the average profile occupancy in control period t is:

$$\bar{o}_{k,t}^{(i)} = \frac{1}{LT} \sum_{l=1}^L \sum_{m=1}^M \bar{o}_{l,m,k,t}^{(i)}, \quad (1)$$

where l is the index of lane, m is the index of aggregation period of detectors, L and M are the number of lanes and aggregation periods in a control period respectively. The traffic state of a segment in control period t is defined as a vector $\mathbf{o}_t^{(i)} = (\bar{o}_1^{(i)}, \bar{o}_2^{(i)}, \dots, \bar{o}_K^{(i)})$, where K is the number of detection profiles which include a group of loop detectors each. We recommend that K is at least 4 to depict the traffic state comprehensively. Also, both upstream and downstream (especially bottleneck area) should be covered by detector groups.

In multi-agent ramp metering tasks, the reward values ($r_t^{(i-1)}$, $r_t^{(i+1)}$ or both) calculated by the neighboring agent(s) can be appended into the state vector, which will be introduced below.

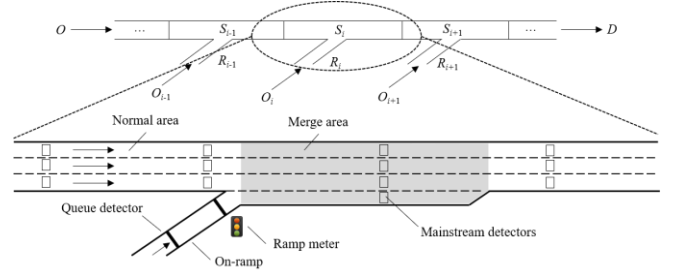


Figure 1. Layout of a freeway section.

2) Action

Consistent with most practical ramp metering operations, the proposed system adopts a one-car-per-green realization. In such setting, a constant-duration green phase permits exactly one vehicle to pass, thus, the ramp volume is controlled by varying the red-phase duration between a minimum and a maximum value. So, the action determined by the agent is a metering rate $MR_t^{(i)}$ calculated based on the observation of state, then the red-phase duration is set as:

$$a_t^{(i)} = \begin{cases} a_{\max} & MR_t^{(i)} < MR_{\min} \\ a_{\min} & MR_t^{(i)} > MR_{\max} \\ (1/MR_t^{(i)} - 1)g_t^{(i)} & \text{otherwise} \end{cases}, \quad (2)$$

where $a_t^{(i)}$ denotes the red-phase duration and $g_t^{(i)}$ denotes green-phase duration. The metering rate will be re-evaluated at the end of each control period. Each agent only need to determine one metering rate of the corresponding ramp in multi-ramp scenarios under the decentralized framework, which can significantly reduce the computational burden.

3) Reward Function

Selection of the reward function has significant impact on the eventual policy arrived at. The objective of ramp metering is to prevent the downstream traffic flow from potential breakdown, i.e., to maintain the travel speed in the merge area. It is also necessary to keep the on-ramp queue below the storage capacity limit, thus preventing long queues to interfere with surface street traffic. Hence, the delayed reward function is designed considering the trade-off between (i) downstream speed and (ii) queue size on the ramp, denoted as:

$$r_{t+1}^{(i)} = \bar{v}_{k_{\zeta},t}^{(i)} - \eta \bar{q}_t^{(i)}, \quad (3)$$

where $\bar{v}_{k_{\zeta},t}^{(i)}$ represents the average speed of profile k_{ζ} in the merge area and $\bar{q}_t^{(i)}$ represents the average queue size on the ramp i in the last control period. The coefficient η reflects the stress on queue size: a larger η encourages eliminating queue on the ramp. The adopted value of η is 0.1 due to the good performance based on the parameter searching process. In multi-ramp scenarios, the reward value can be shared with other agents to achieve coordinated control.

B. Policy Parameterization for Continuous Action Space

Ramp metering problem differs from Atari video games (20) in that the action, i.e. metering rate, can be continuous real numbers rather than discrete keystrokes. Furthermore, the value of metering rate is interval variable, thus a straightforward discretization leads to the loss of the scale information and makes the learning progress slow. Therefore, the agents should learn a continuous probability distribution function (or a parameterized policy) instead of many discrete actions.

To produce a policy parameterization, the policy is defined as the normal probability density over a real-valued scalar action, with mean and standard deviation given by parametric functions that depend on states, denoted as:

$$\pi_{\theta}(a^{(i)} | o_t^{(i)}) = \frac{1}{\sigma_{\theta}(o_t^{(i)})\sqrt{2\pi}} \exp\left(-\frac{(a^{(i)} - \mu_{\theta}(o_t^{(i)}))^2}{2\sigma_{\theta}(o_t^{(i)})^2}\right), \quad (4)$$

where μ and σ are parameterized function approximators for mean and standard deviation respectively, and $\theta = [\theta_{\mu}, \theta_{\sigma}]^T$ is parameter vector.

C. Proximal Policy Optimization

In the present paper, a newly proposed RL algorithm, Proximal Policy Optimization (PPO) [16], [17], is adopted to get a desirable ramp metering policy. The algorithm adopts a classical Actor-Critic (AC) framework which combines the strong points of policy-based and value-based methods as shown in Section 2.2, and imports the Kullback–Leibler (KL) divergence to adjust the update rate of the policy function automatically for good adaptability.

The backend neural network used in this paper is demonstrated in Fig. 2. Actor network is a value function approximator structured as a simple fully connected neural network. It takes a state-action pair $(o_t^{(i)}, a_t^{(i)})$ as input and estimates its value. The loss function is the square error:

$$L_{SE}(\varphi) = -\sum_{i=1}^T \left(\sum_{t'=t}^T \gamma^{t'-t} r_{t'}^{(i)} - V_{\varphi}(o_t^{(i)}) \right)^2, \quad (5)$$

where φ is parameters of value function. Critic network takes a state-action pair $(o_t^{(i)}, a_t^{(i)})$ as input and estimates μ and σ respectively. The update rule takes use of a specially designed PPO error function, which will be introduced below. Adam [18], a first-order gradient-based optimization approach is used for weight update of the neural networks.

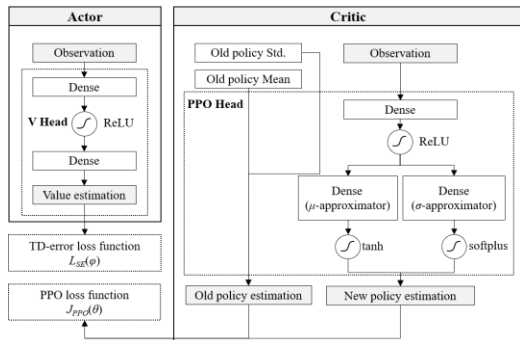


Figure 2. Network structure used in the algorithm.

The following tricks are embedded in the algorithm:

(1) *Advantage calculation* The term *advantage* refers to the difference value between the action value Q and the state value V [19]. Since we cannot determine the Q value directly while training, we can use the return, i.e., the discounted sum of rewards, as an estimation of Q . Thus, the advantage is:

$$A_t = \sum_{i=t}^T \gamma^i r_i^{(i)} - V_{\varphi}(o_t^{(i)}), \quad (6)$$

The advantage allows the agent to determine how much better they turned out to be than expected. We hope the value is neither too large nor too small to make the training process smooth.

(2) *PPO loss function* The actor network of PPO adopts a specified loss function, denoted as:

$$J_{PPO}(\theta) = \sum_{i=1}^T \frac{\pi_{\theta}(a_t^{(i)} | o_t^{(i)})}{\pi_{old}(a_t^{(i)} | o_t^{(i)})} A_t - \lambda KL[\pi_{old} | \pi_{\theta}], \quad (7)$$

where $\lambda KL[\pi_{old} | \pi_{\theta}]$ is a regularization term, in which $KL[\pi_{old} | \pi_{\theta}]$ represents the KL divergence between the policies π_{old} and π_{θ} defined as:

$$KL[\pi_{old} | \pi_{\theta}] = \int_{-\infty}^{\infty} \pi_{old} \log \frac{\pi_{old}}{\pi_{\theta}} d\theta, \quad (8)$$

which reflects the similarity between the two probability distributions; λ is the adaptive coefficient. The regularization term will limit the update size of the policy function to avoid divergence [16].

(3) *Distributed implementation* For good performance and high training efficiency, the algorithm can be implemented in a distributed manner. Data collection and gradient calculation are distributed over workers while global parameters of actor and critic network are updated by the chief processing. Distributed implementation was proved to be more effective in continuous action spaces and have better convergence properties [20].

D. Description of the Algorithm

In this subsection we combine all things together to give a complete description of the ramp-metering algorithm based on multi-agent DRL. The algorithm encompasses two stages, i.e. training and inference. In the training stage, the RL agents learn a metering policy through interaction with the environment; In the inference stage, agents observe the environment then take proper actions given the learned policy. Inference only use the μ -approximator to make decisions. The agents can share the delayed reward with others for collaboration in both of the two stages. In addition, pre-trained networks can be transferred to other freeway ramp metering scenarios with only slight modification, e.g., changing the number of input nodes. The following algorithm box shows the complete algorithm. It should be noted that though the training process requires multi-core computation platform to achieve distributed training, the inference stage needs far less computing resource, which makes the deployment of algorithm tractable.

Algorithm: Ramp Metering based on MAPPO Framework

I. Training Stage

Initialization: Initialize the actor and critic network with random parameters;

For each episode do:

Reload the ramp metering environment;

Run policy $\pi_{\theta}^{(i)}$ for T timesteps, collecting $(o_t^{(i)}, a_t^{(i)}, r_{t+1}^{(i)})$;

Estimate advantage $A_t = \sum_{t' > t} \gamma^{t'-t} r_{t'}^{(i)} - V_{\phi}^{\pi}(o_t^{(i)})$;

$\pi_{old}^{(i)} \leftarrow \pi_{\theta}^{(i)}$;

While not done do:

$$J_{PPO}(\theta) = \sum_{t=1}^T \frac{\pi_{\theta}(a_t^{(i)} | o_t^{(i)})}{\pi_{old}(a_t^{(i)} | o_t^{(i)})} A_t - \lambda KL[\pi_{old} | \pi_{\theta}];$$

Update θ by Adam optimization algorithm;

While not done do:

$$L_{SE}(\phi) = -\sum_{t=1}^T \left(\sum_{t' > t} \gamma^{t'-t} r_{t'}^{(i)} - V_{\phi}^{\pi}(o_t^{(i)}) \right)^2;$$

Update ϕ by Adam optimization algorithm;

If $KL[\pi_{old} | \pi_{\theta}] > KL_{max}$ **then:**

$\lambda \leftarrow \alpha \lambda$;

Else:

$\lambda \leftarrow \lambda / \alpha$;

II. Inference Stage

Initialization: Load the parameters of the critic (policy) network;

Loop forever:

Observe the state $o_t^{(i)}$;

Append the reward(s) of other agents to the state:

$o_t^{(i)} \leftarrow o_t^{(i)} \cup r_t^{(j)}$;

Output the metering action $a_t^{(i)}$ using the μ -approximator;

Calculate the reward $r_{t+1}^{(j)}$;

III. SIMULATION EXPERIMENTS

To evaluate the performance of the proposed algorithm, simulation experiments with various freeway network and traffic demand scenarios are conducted based on a customized simulation platform.

A. Simulation Platform

The proposed DRL-based ramp metering algorithm reach good performance by training neural network policies, which requires repeated interaction with the environment. We build up a simulation platform named ST4RLRM (an acronym for the Simulation Testbed for Reinforcement Learning Ramp Metering). The work is mainly inspired by a computational framework for deep RL and traffic control experiments *Flow* [21]. An opensource road traffic simulator SUMO (Simulation of Urban Mobility) [22] is adopted as the simulation backend. Ramp metering logic is implemented in Python script language and TensorFlow machine learning framework.

The components and workflow of ST4RLRM platform are demonstrated in **Fig. 4**. Given the traffic demand profile based on the historical observation, as well as the freeway network, the backend traffic simulator will run the simulation and record specified measurements (i.e., occupancy) as input of ramp metering algorithms. Algorithmic modules then take the measurements to determine the metering rate in real time. The algorithm library contains a set of ramp metering algorithms for convenient test and comparison. As to the training of the proposed PPO-based algorithm, SUMO has the capability to run multiple simulation instances parallelly. A well-configured algorithm will be taken into implementation.

During the implementation, the training of the PPO agents can still be in progress in back end. The only difference lies in that the measurements should be collected from the realistic traffic environment. The feature makes self-improvement of the metering agent possible.

All of our following experiments were carried out on a computing server with one i7-8700K CPU (3.7GHz, 6 Cores). There is no need of dedicated GPU for simulation or training.

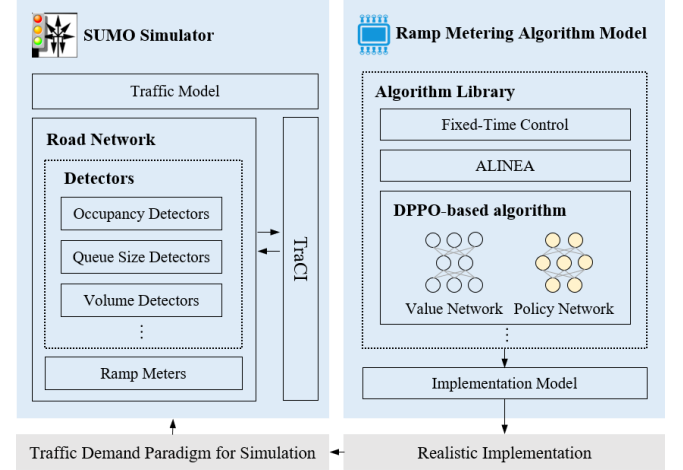


Figure 3. Schematic view of ST4RLRM platform.

B. Experiment Design

1) Simulation Set-up

Various types of network structures and traffic demand profiles are required to check out the efficiency of the proposed algorithm. We built up both sing-ramp and multi-ramp simulation scenarios (**Fig. 5**). The single-ramp scenario embodies freeway mainline which is about 1km in length and one on-ramp. The multi-ramp scenario contains mainline which length of 1.7km and two on-ramps. Loop detectors are set on mainline every 200 meters and queue size detectors are equipped on each on-ramp. We use the occupancies of four profiles near the ramp as the representation of state and the reward is calculated based on the profile speed of bottleneck point and ramp queue size (see (11)). Three types of demand profiles: (i) stationary volume (**Fig. 6** (a, d)), (ii) flat peak (**Fig. 6** (b, e)), and (iii) sharp peak (**Fig. 6** (c, f)), are designed to simulate different traffic conditions for mainline (**Fig. 6** (a-c)) and on-ramp (**Fig. 6** (d-f)). We use the flat peak profile in the training stage. To avoid overfitting, a small random noise is appended to the profile before each training episode.

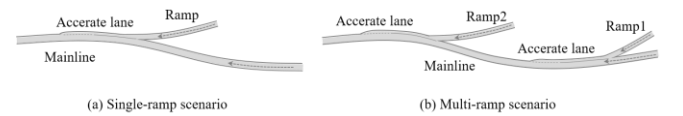


Figure 4. Freeway network used in experiments.

As shown in **Fig. 6**, one simulation (episode) for training and inference is set as 2 hours. The control period is set as 1 minute. Hence, an episode will encompass 120 control periods. One-car-per-green mode is adopted; thus, one cycle

includes 3 seconds green phase followed by a variable duration red phase. A new control period starts only when the last cycle ends.

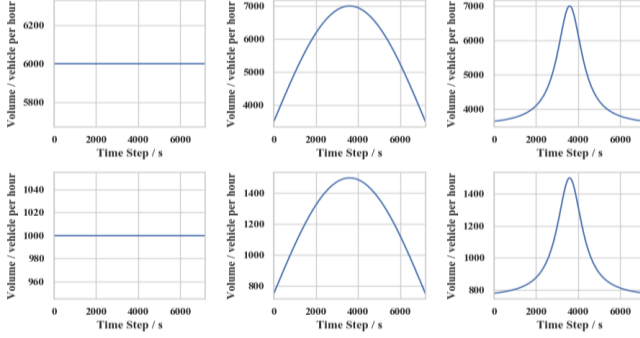


Figure 5. Demand profiles of mainline and ramp.

2) Tested Algorithms

Besides the proposed algorithm, other metering strategies are tested in our simulation experiments. The state-of-the-art ramp metering algorithm ALINEA is used as benchmark. Grid search has been done to find the best parameter combination of ALINEA. In all of the experiments we set the critical occupancy to 0.18 and the metering gain KR to 0.35 (as recommended in [23]). No-control and fixed-time control are also tested for comparison. The proper metering rates of fixed-time control are determined by enumeration experiment in different scenarios.

C. Experiment Result

1) Single-Ramp Scenario

We finished the training process of single-ramp scenario when the episode number reached 1000. The elapsed time is less than 6 hours in our training platform. The training process is presented in **Fig. 7**. It can be seen that the agent made poor choices in the initial stage, but then made rapid progress soon; following a rising process the return tends to convergence after 400 episodes. The average speed at the bottleneck point increased about 8.6% and the average queue size on the ramp decreased a lot from the initial worst condition.

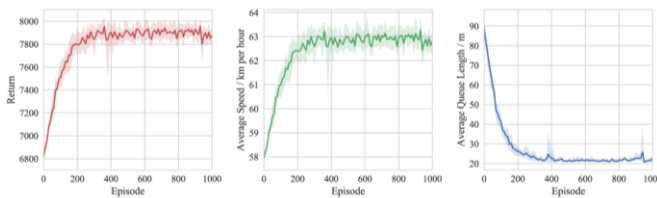


Figure 6. Log visualization of the training process.

The policy network is thrown into utilization after training process. Four types of ramp metering approaches, i.e. (i) no-control, (ii) fixed-time control, (iii) ALINEA and (iv) the proposed algorithm are tested in three demand scenarios. The result is presented in **Table I**. No-control strategy always leads to the minimum queue size on ramp but gets the lowest mainstream speed. Fixed-time strategy ameliorates mainstream traffic somehow, but the effect is limited. The

performance of ALINEA is unstable. In stationary volume scenario, ALINEA reaches the highest speed but produces overlong queue size. In flat peak scenario, ALINEA gets worse performance than fixed-time control. The proposed algorithm achieves fairly desirable performance and gets the highest return under all of the demand profiles.

TABLE I. COMPARISON OF METERING STRATEGIES

Demand Profile	Metering Strategy	Measures		
		Speed (km/h)	Queue size (m)	Return
Stationary Volume	No-control	38.62	4.57	4579.59
	Fixed-time	47.84	39.49	5266.78
	ALINEA	57.31	118.81	5450.99
	Proposed	55.24	92.26	5521.68
Flat Peak	No-control	43.08	4.45	5116.61
	Fixed-time	51.49	37.28	5731.62
	ALINEA	49.89	38.74	5522.56
	Proposed	55.71	28.32	6345.36
Sharp Peak	No-control	54.66	1.47	6541.25
	Fixed-time	59.76	38.37	6711.21
	ALINEA	57.54	11.09	6771.72
	Proposed	60.97	29.34	6964.29

Shown in **Fig. 8** is the space-time diagram of mainstream speed under different metering strategies. Inset (a-d), (e-h), (i-l) show the results of stationary volume, flat peak and sharp peak demand scenario embedded with no-control, fixed-time control, ALINEA and the proposed algorithm respectively. It is clearly that the proposed algorithm can decrease the duration as well as the spatial range of mainline congestion in a variety of demand scenarios. Since the policy network had not been trained in all of these scenarios specifically, the result also presents the scalability and robustness of the algorithm.

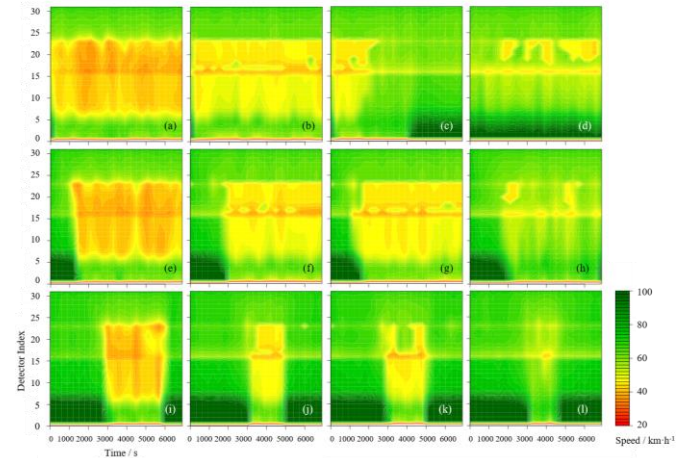


Figure 7. Space-time diagram of speed under different metering strategies.

2) Multi-Ramp Scenario

In multi-ramp scenario, though the metering agents make decisions respectively, they are capable of utilizing reward information of other agents to achieve collaborative control. **Fig. 9** demonstrates the performance (reward, mainline speed and ramp queue size in a 2-hour episode) of the proposed

algorithm in a double-ramp network like **Fig. 5 (b)** in flat peak demand profile. Compared to the benchmark ALINEA controller, the proposed algorithm keeps a higher mainline speed in average, and prevents the queue length from extending. Furtherly, no huge speed difference is produced by the proposed algorithm, which benefits homogeneity of traffic flow.

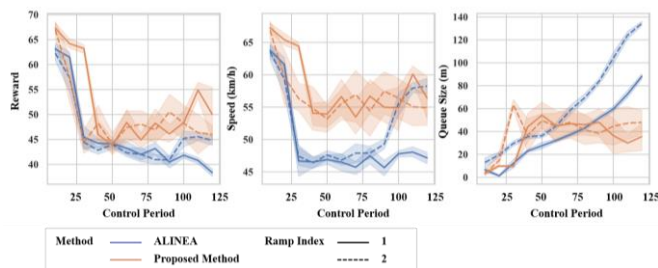


Figure 8. Control performance of ALINEA and proposed algorithm.

IV. CONCLUSION

The advance of reinforcement learning empowers the accurate control of stochastic systems. In this paper, we proposed a novel ramp metering algorithm, in which the multi-agent deep reinforcement learning technique is the primary enabler. The algorithm works in a model-free fashion: the agents take the profile occupancies as input and output the metering rates directly (end to end), without the requirement to pre-assume a freeway traffic flow model. The metering approach is modelled as a policy neural network. Minimal fine-tuning is required when the implementation environment is changed. Hence, the algorithm provides favorable flexibility for time-varying demand and migrating implementation.

We developed a SUMO-based platform to implement the algorithm. A set of simulation experiments demonstrated that the DRL-based ramp metering method is able to automatically smooth traffic flow, thus postponing the occurrence and mitigating the impact of congestion at the bottleneck. In the single-ramp experiments, the performance of the proposed algorithm is 1.3%, 14.9% and 2.8% higher than the state-of-the-practice ramp metering algorithm ALINEA in the stationary, flat peak and sharp peak demand scenarios, respectively. In the multi-ramp scenario, our algorithm also presents the capability of keeping mainstream speed at the same level and preventing the spread of queuing on the ramp, especially during the occurrence of long-lasting congestion.

In this work, a relatively simple reward function is adopted. For future work, the effects of different reward definitions in the context of ramp metering need to be carefully investigated. In addition, the form of cooperation in multi-agent scenarios is worth exploring as well. Finally, a recurrent neural network with predictive mechanism for forward-looking control is suggested to improve the performance of control systems.

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